

Artifact Comparator Developer Manual

1. How to Obtain Source Code

- **Use a modern OS:** Windows 10/11, macOS Monterey+, or a recent Linux distro.
- **Install:** Node.js 18+, npm 9+, Git, and a modern browser.
- **Clone the repo:**
 - `git clone https://github.com/CS319-25-FA/S1-T2-Databaseable.git`
 - `cd artifact-comparator`

2. Layout Of Directory Structure

S1-T2-Databaseable/

```

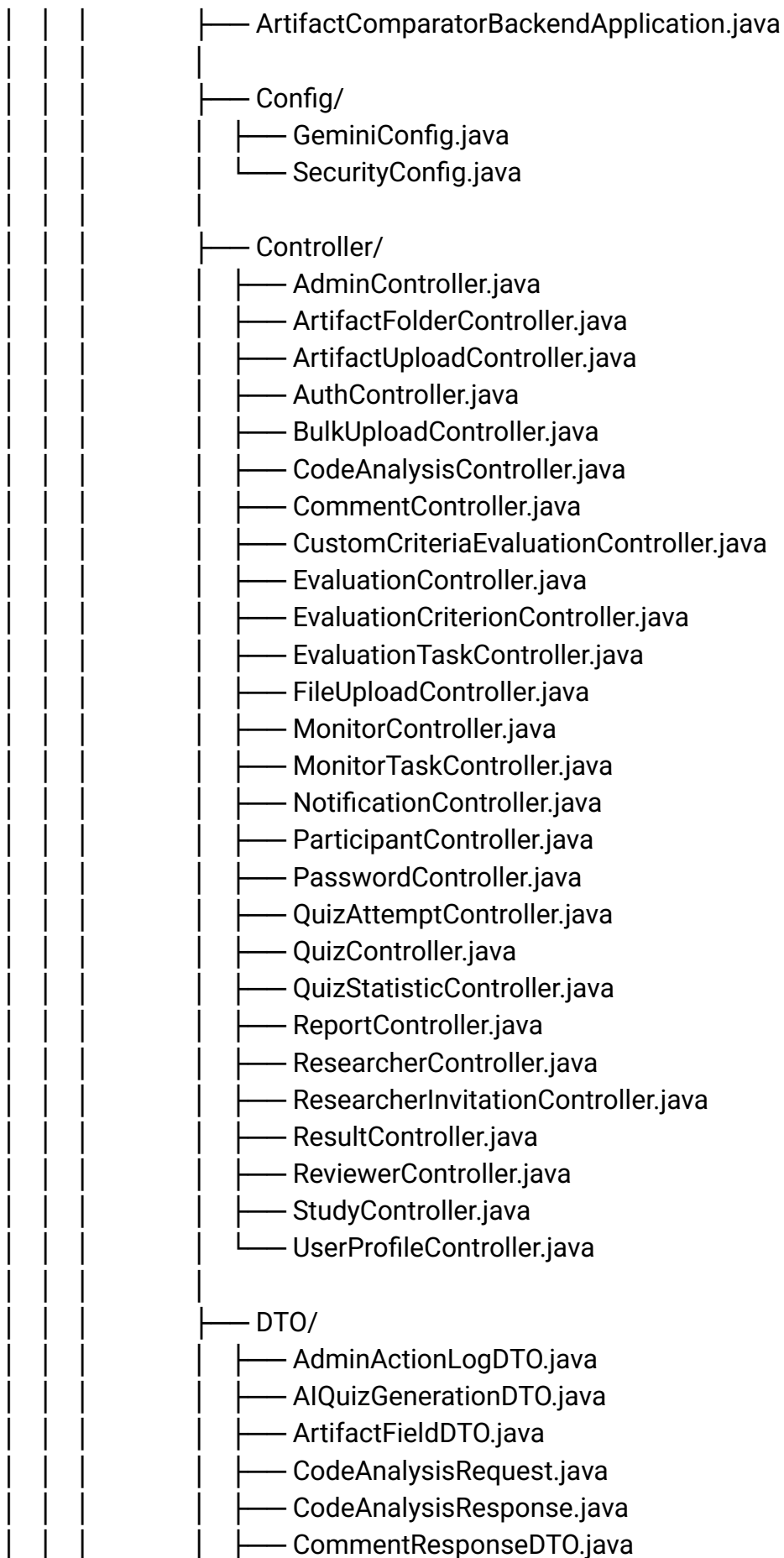
├── .idea/                                # IntelliJ IDEA configuration
├── backend/ [artifact-comparator-backend] # Spring Boot Backend
├── frontend/                             # React Frontend
├── deliverables/                         # Project deliverables
│   └── S1T2-D5/
│       ├── S1T2-D0.pdf
│       ├── S1T2-D1.pdf
│       ├── S1T2-D2.pdf
│       └── S1T2-D3.pdf
├── docker-compose.yml                    # Multi-container orchestration
└── README.md

```

Backend Structure

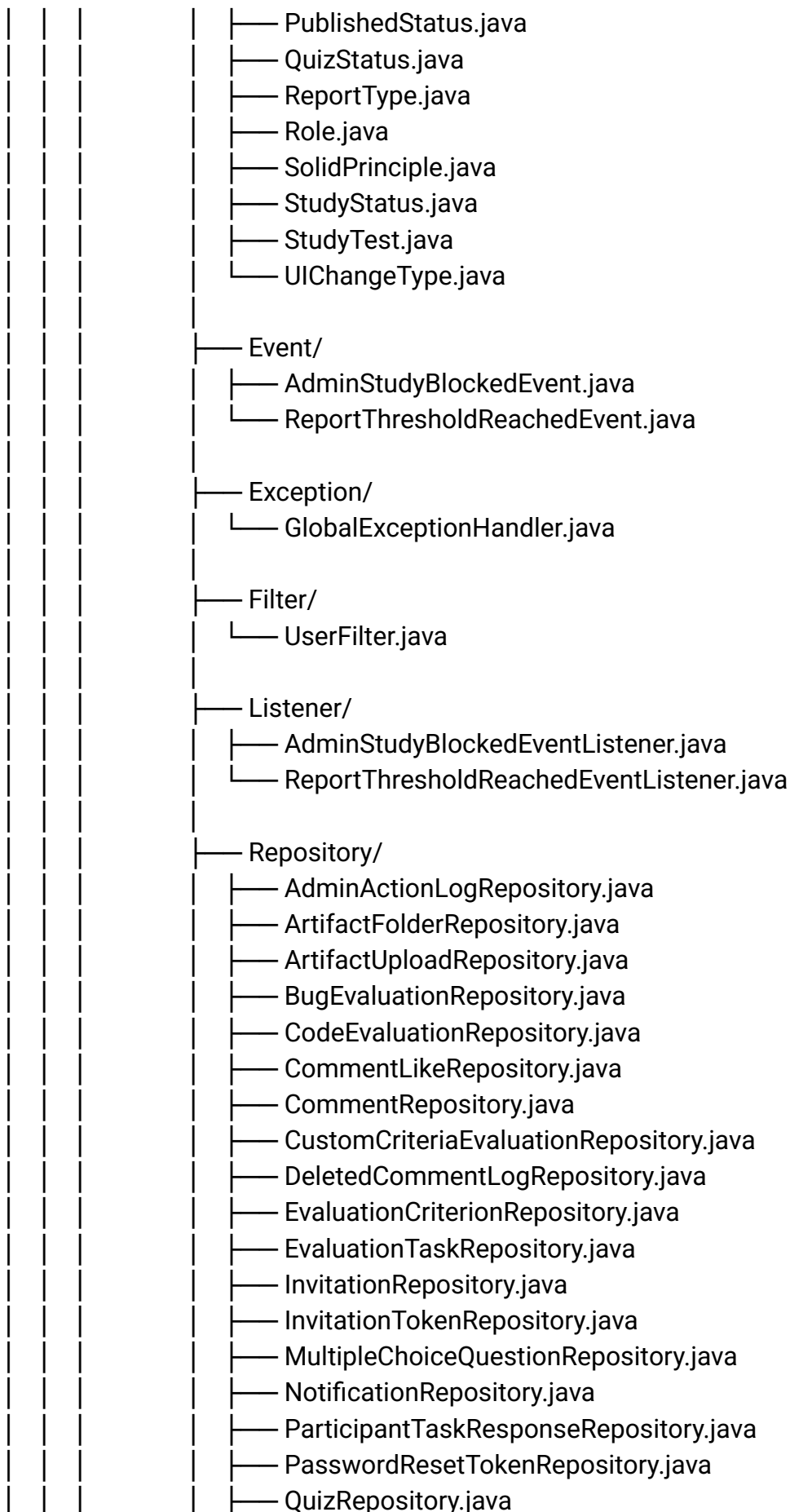
backend/ [artifact-comparator-backend]

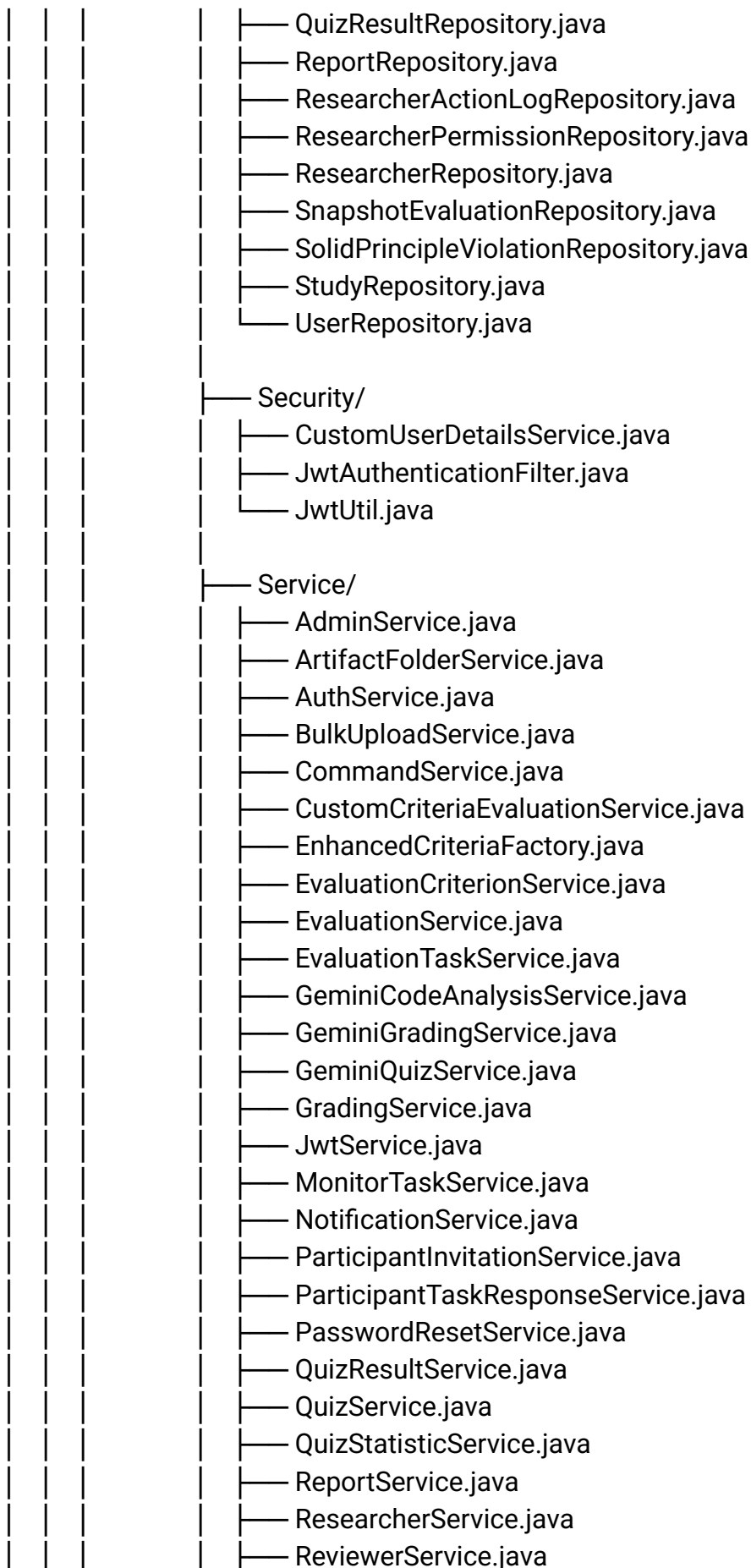
```
|— .idea/                                # IntelliJ IDEA configuration
|— src/
|   |— main/
|   |   |— java/
|   |   |   |— com/
|   |   |   |   |— artifactcomparator/
|   |   |   |   |   |— artifact_comparator_backend/
```



				ComplexityMetric.java
				CreateCommentRequestDTO.java
				CreateCriterionRequest.java
				CreateReportRequestDTO.java
				CustomCriteriaSubmitRequestDTO.java
				DeletedCommentLogDTO.java
				EvaluationRequestDTO.java
				EvaluationTaskCreateRequest.java
				ForgotPasswordDTO.java
				GradingResult.java
				InviteResearcherRequestDTO.java
				LoginDTO.java
				ManualGradingRequestDTO.java
				MonitorTaskDetailDTO.java
				MultipleChoiceQuestionDTO.java
				NotificationDto.java
				OpenEndedQuestionDTO.java
				ParticipantProgressDTO.java
				ParticipantStudyProgressDTO.java
				PasswordChangeDTO.java
				QuestionDTO.java
				QuizAttemptResponseDTO.java
				QuizCreateDTO.java
				QuizResultDTO.java
				QuizSubmissionDTO.java
				QuizUpdateDTO.java
				RegisterDTO.java
				ReportResponseDTO.java
				ResearcherActionLogDTO.java
				ResearcherPermissionDTO.java
				ResearcherResponseDTO.java
				ResetPasswordDTO.java
				SolidPrincipleDTO.java
				StudyMonitorDTO.java
				StudyRequestDTO.java
				StudyResponseDTO.java
				TaskDetailsDTO.java
				TaskMonitorDTO.java
				UpdateStudyRequestDTO.java
			Entity/	
				AdminActionLog.java

				ArtifactFolder.java
				ArtifactUpload.java
				BugEvaluation.java
				CodeEvaluation.java
				CodeRefactorRevision.java
				Comment.java
				CommentLike.java
				CorrectAnswerEntity.java
				CriteriaUser.java
				CustomCriteriaEvaluation.java
				DeletedCommentLog.java
				EvaluationCriterion.java
				EvaluationTask.java
				ImageHighlightCitation.java
				Invitation.java
				InvitationToken.java
				MultipleChoiceCriterion.java
				MultipleChoiceQuestion.java
				Notification.java
				OpenEndedCriterion.java
				OpenEndedNumericCriterion.java
				OpenEndedQuestion.java
				ParticipantTaskResponse.java
				PasswordResetToken.java
				Question.java
				Quiz.java
				QuizResult.java
				RatingCriterion.java
				Report.java
				Researcher.java
				ResearcherActionLog.java
				ResearcherPermission.java
				SnapshotEvaluationRepository.java
				SolidPrincipleViolation.java
				Study.java
				User.java
			Enums/	
				BugCategory.java
				CodeType.java
				CriterionType.java
				NotificationType.java





```

|
|
|
|
|   |   |   |   |
|   |   |   |   |--- StudyInviteExportService.java
|   |   |   |   |--- StudyService.java
|   |   |   |   |--- TaskArtifactService.java
|   |   |   |   |--- UserService.java
|   |   |   |
|   |   |   |--- Util/
|   |   |   |   |--- CdnResourceUtil.java
|   |   |   |   |--- ZipUtil.java
|   |   |
|   |   |--- resources/
|   |   |   |--- application.properties
|   |   |   |--- (other resource files)
|   |
|   |--- test/
|   |   |--- java/
|   |   |   |--- com/
|   |   |   |   |--- artifactcomparator/
|   |   |   |   |   |--- artifact_comparator_backend/
|   |   |   |   |   |--- ArtifactComparatorBackendApplicationTests.java
|
|--- target/                # Maven build output
|--- .mvn/                  # Maven wrapper
|--- HELP.md                # Spring Boot help
|--- mvnw                   # Maven wrapper (Unix)
|--- mvnw.cmd               # Maven wrapper (Windows)
|--- pom.xml                # Maven dependencies
|--- README.md              # Backend documentation

```

Frontend Structure

```

frontend/
|--- node_modules/          # npm dependencies (library root)
|--- public/
|   |--- index.html
|   |--- favicon.ico
|   |--- manifest.json
|   |--- robots.txt
|--- src/
|   |--- components/
|   |   |--- AddTaskPage.js
|   |   |--- AdminIndex.js
|   |   |--- ArtifactDetailPage.js

```

- BulkUploadPage.js
- CreateQuizPage.js
- CreateStudyPage.js
- DefineStudyTemplatePage.js
- EditQuizPage.js
- EvaluationTasksPage.js
- ForgotPassword.js
- InformationPage.js
- InvitationService.js
- JoinStudyPage.js
- Login.js
- ManageArtifactsForTaskPage.js
- ManageCorrectAnswerPage.js
- ManageEvaluationCriteriaPage.js
- ManageEvaluationTasksPage.js
- ManageParticipantsPage.js
- ManageResearchersPage.js
- ManageSingleTaskPage.js
- ManageStudyPage.js
- MonitorEachTaskPage.js
- MonitorStudyProgressPage.js
- MyArtifactsPage.js
- MyInvitationsPage.js
- MyStudiesPage.js
- NotificationWidget.js
- ParticipantIndexPage.js
- PendingQuizReviewPage.js
- ProfilePage.js
- ProtectedRoute.jsx
- QuizManagementPage.js
- QuizResultPage.js
- QuizStatisticsPage.js
- reportWebVitals.js
- ResearcherIndexPage.js
- ResearcherReviewDashboard.js
- ResetPassword.js
- ReviewerIndexPage.js
- setupTests.js
- SignUp.js
- TakeQuizPage.js
- TaskCommentsPage.js
- TaskDetailsPage.js

			TaskReportsPage.js	
			UploadArtifactPage.js	
			App.css	
			App.js	
			App.test.js	
			index.css	
			index.js	
			logo.svg	
			uploads/	# User uploaded files
			.gitignore	# Git ignore rules
			docker-compose.yml	# Frontend Docker compose
			package.json	# npm dependencies
			package-lock.json	# npm lock file
			Dockerfile	# Frontend Docker image
			README.md	# Frontend documentation

3. How to Build Software

- Install and run frontend:
 - `docker compose up`
 - `cd artifact-comparator`
 - `npm install`
 - `npm start (dev)` or `npm run build` then serve the build/ folder.

```

• spring.application.name=Artifact Comparator Backend
•
• # --- PostgreSQL Connection ---
• spring.datasource.url=jdbc:postgresql://localhost:5432/artifact_db
• spring.datasource.username=postgres
• spring.datasource.password=postgres
•
• # --- Hibernate / JPA Settings ---
• spring.jpa.hibernate.ddl-auto=update
• spring.jpa.show-sql=true
• spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.PostgreSQLDialect
• spring.jpa.properties.hibernate.format_sql=true
•
• # --- Multipart (File Uploads) ---

```

```
• spring.servlet.multipart.max-file-size=200MB
• spring.servlet.multipart.max-request-size=200MB
•
• spring.mail.host=smtp.gmail.com
• spring.mail.port=587
• spring.mail.username=artifactcomparator@gmail.com
• spring.mail.password=*****"Insert your password here"
• spring.mail.properties.mail.smtp.auth=true
• spring.mail.properties.mail.smtp.starttls.enable=true
•
• # --- NEW: Gemini AI Configuration ---
• gemini.api.key=*****
• gemini.api.url=https://generativelanguage.googleapis.com/
  v1beta/models/gemini-2.5-flash:generateContent
• gemini.api.timeout=30000
```